Lab04

Task 1 (1 point):

Open your GitHub repository in Codespace.

1. In the terminal run the command: `cd Labs/Lab04/Employee/`
2. In the terminal run the command: `cp ../../Lab03/University/.env .env`
3. In the terminal run the command: `mkdir DATA/`
4. In the terminal run the command: `touch DATA/employee.csv DATA/company.csv DATA/works.csv`
5. In the terminal run the command: `touch ddl.sql main.py`

Task 2 (3 points):

In the file called **ddl.sql**, provide the create table statements for the following schema:

employee (id, person_name, street, city, state)
company (company_name, city, state)
works (id, company_name, salary)

- Provide primary keys and foreign keys.
- Only salary should be numeric. All other attributes should be varchar.
- All attributes should reject null values, except works' attributes. (Primary keys reject null values by default.)
- Ensure the id attribute is exactly 7 characters longs.
- Ensure state is exactly 2 characters long.
- Ensure salary is no less than $40,000 and no greater than $200,000.

Task 3 (3 points):

In each of the CSV files in **DATA/**, add at least 6 records that can later be loaded into the database.

Task 4 (3 points):

In the file called main.py, add the following to the top of the file:

```
from setup import connect_to_db, create_database, create_tables
from load import load_data, show_tables, get_relation_names
from drop import drop_tables
```

1. Create a variable that holds the connection to the database.
2. If the connection is not None, call the create_database() function with the connection variable and "employee_db" as arguments, then close the connection.
3. Create a variable that holds the connection to the newly created database by passing "employee_db" as an argument.
4. If the connection is not None:
    a. Call the create_tables() function with the connection variable as an argument.
    b. Create a variable that holds the relation names from calling get_relation_names()
    c. Load the database with connection and the relations variables as arguments.
    d. Show all tables.
    e. Close the connection.
5. Run **main.py**.

Push your work to GitHub using the following commands:

```
git pull
git add .
git commit –m "lab04 upload"
git push
```